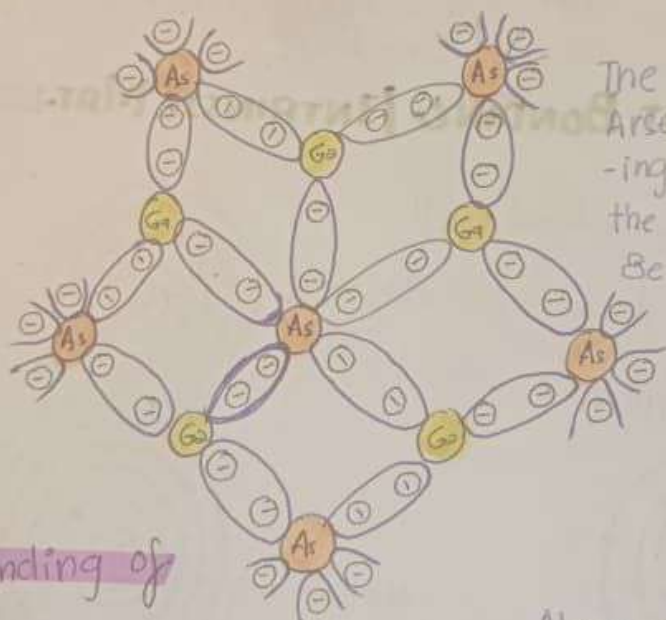




Covalent Bonding of the Silicon Atom -

The four valence shells forming bond with other Si atoms to complete the outermost shell of 8 electrons.



The Gallium and Arsenic are forming bonds to form the complete shell of 8.

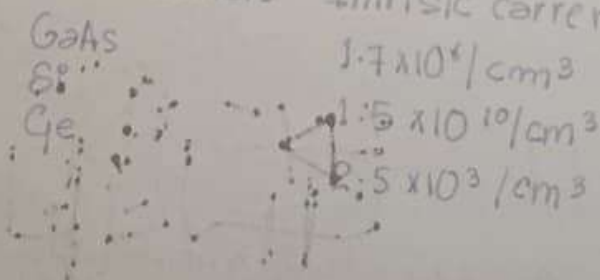
Covalent Bonding of GaAs crystal

As it is a compound semiconductor, there is sharing b/w two different atoms.

At room temp. there are 1.5×10^{10} free charge carriers in intrinsic Si (1 cm^3).

The term intrinsic applied to a semiconductor that has been refined to reduce the number of impurities (purest).

Semiconductors	Intrinsic carriers
GaAs	$1.7 \times 10^6 / \text{cm}^3$
Si	$1.5 \times 10^{10} / \text{cm}^3$
Ge	$2.5 \times 10^{13} / \text{cm}^3$



SEMICONDUCTOR MATERIALS

Group 14 (IV)

Carbon (C)

Gallium Arsenide (GaAs) — Gallium, Arsenic

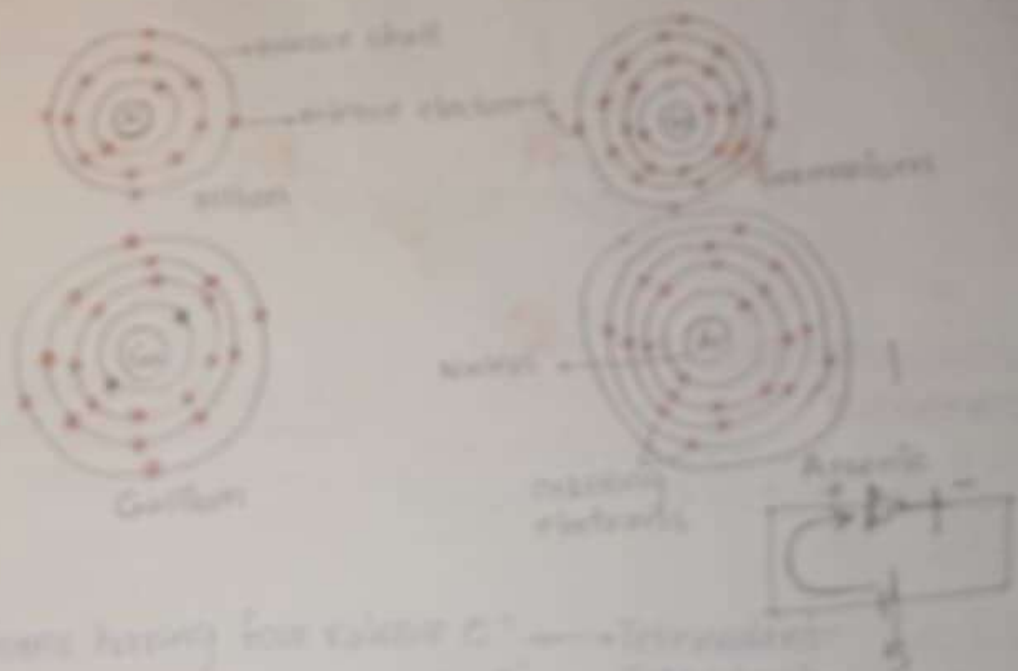
Semiconductors are materials having conductivity between that of a good insulator and an conductor.

Semiconductors are classified into two groups.

(i) Direct Bandgap — have valence band and conduction band.

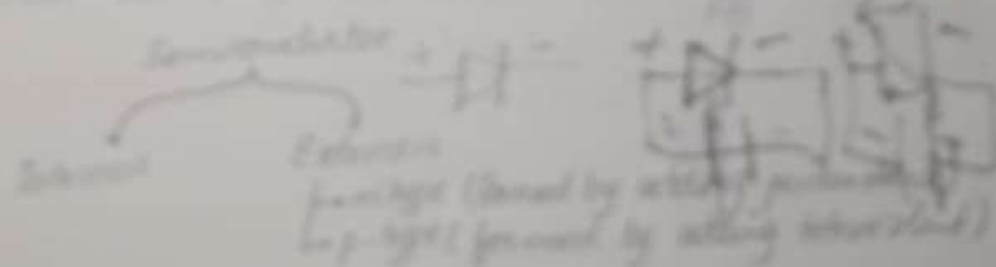
(ii) Indirect Bandgap — have valence band and conduction band. The conduction band is not at the minimum of the valence band.

COVALENT BONDING & INTRINSIC MRS.



Atoms having four valence electrons —> Tetra-valent
 Atom having four valence electrons —> Tetra-valent
 (i.e. 4 valence electrons)

From "valence" 4, it is called as "tetra-valent"



n-type & p-type materials

Semiconductor material that has been gone through the doping process is called extrinsic material.

There are two extrinsic materials

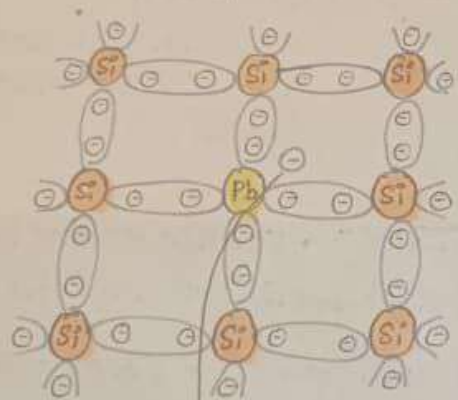
- n-type materials.
- p-type materials.

{ formed by adding a
predetermined no.
of impurities in Si }

n-type materials

- created by introducing impurity elements, that have five electrons (pentavalent)-

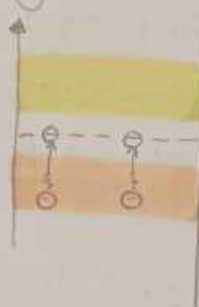
- antimony
- arsenic
- phosphorus



Fifth valence e^- of antimony-

Diffused impurities with 5 valence e^- are donors-

four e^- of Si will form a bond of e^- from antimony - Hence, the remaining e^- after the completion of shell will "free" to move within the newly formed n-type material-



E_g = relatively lesser than of the purest element-

Extrinsic:-

- pure semiconductor material. lower conductivity. At 0K, no free charges are available. upon given energy, it has equal number of electrons and holes (absence of electrons in valence shell). negative temperature coefficient-

Extrinsic:-

- It has higher conductivity. Impurities will define number of e^- and holes. Impurities are added in ppm. N-type & P-type.
- Majority carriers are electrons; minority $\rightarrow$ holes.

Due to the extra number of e^-

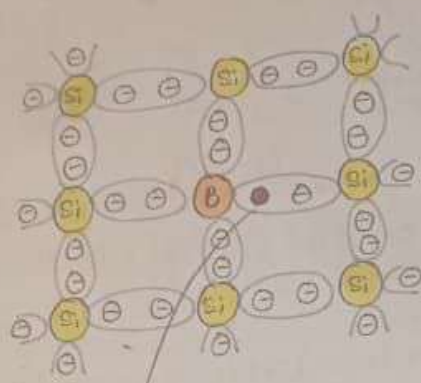
$$5e^- - 4e^- \rightarrow 1e^-$$

$\rightarrow$ in the shell (majority)

$\rightarrow$ Energy level

P-type materials

- Are formed by adding trivalent impurities (one free hole)
- known as acceptor impurity - Aluminium, Boron (III)
- majority carriers $\rightarrow$ holes
- minority carriers $\rightarrow$ electrons

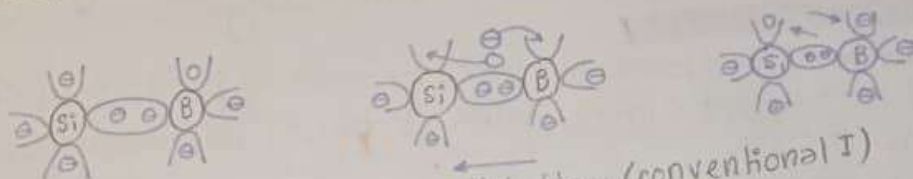


void (0 or +)

The four electrons of Si will form a bond with the three electrons of Boron. This will leave a hole in the valence shell.

Doping is addition of calculated impurities in Si.
intrinsic + doping = extrinsic

ELECTRON VERSUS HOLE flow



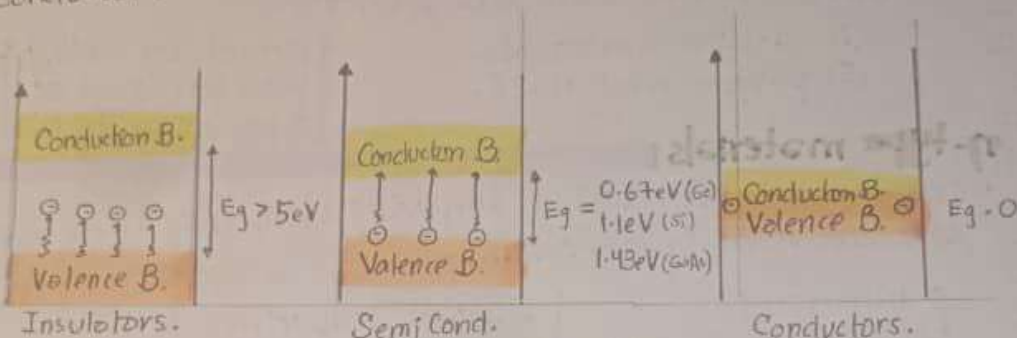
Hole flow (conventional I)
Current flow (natural / real I)

Electron gained KE $\rightarrow$ it will go into vacancy (hole) $\rightarrow$ as electron move to the vacant space $\rightarrow$ A vacancy was created in the place of electron (previous space) $\rightarrow$ This will trigger another electron to fill the vacancy of previous electron $\rightarrow$ Thus, a series would be started of electrons moving to RIGHT and holes moving towards LEFT.

The direction used here is of conventional flow, which is indicated by direction of hole flow.
conventional flow of +ve e $\rightarrow$ the direction of current considered in which the positive charges would move, in that case HOLES.

ENERGY LEVELS

Insulators
Semiconductors
Conductors



Insulators.
Electrons unable to reach conduction B. so no flow of e^- hence, no current flow. E_L is low ≈ 0 .

Semi Cond.
Electrons "free" to establish conduction, hence flow of e^- ; thus the energy levels are moderate.

Conductors.
The bands overlap, has free flow of e^- and thus they have the highest energy levels.

Valence B.

• low energy $\rightarrow$ associated with valence $e^- \rightarrow$ valence e^- are bound in atom. struc, so they (valence e^-) don't participate in I flow.

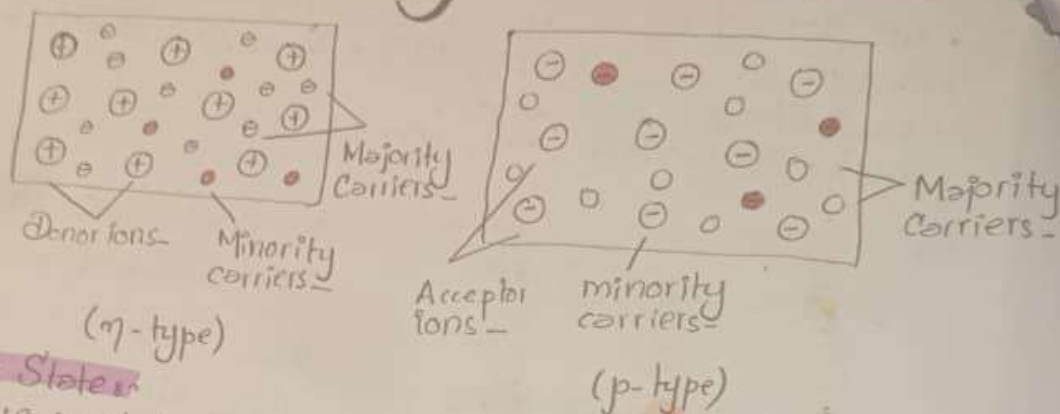
Conduction B.

• higher energy levels $\rightarrow$ associated with free $e^- \rightarrow$ Free e^- aren't bounded in atom. struc, so they (free e^-) only participate in CF.

Forbidden Energy Gap.

• Energy difference b/w the top of the valence band and the bottom of the conduction band.

Majority & Minority Carriers in n -type & p -type materials



Intrinsic State

- holes are created when thermal or light energy breaks cov. b.
- number of electrons = no. of holes.

n -type materials

- Doping with donor atoms (pentavalent) adds extra electrons.
- e^- becomes majority carriers $\rightarrow$ holes remains as minority carriers.

p -type materials

- Doping with acceptor (trivalent) atoms $\rightarrow$ holes are majority charge carriers $\rightarrow$ electrons are minority

Ion Representation

- A donor atom loses electron, becoming positively charged ion $\rightarrow$ accept atom gains electrons, becoming negatively charged ions.

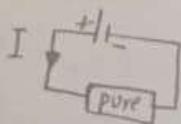
Load Line Analysis

$$e^- = h^+ \text{ (electrons = holes)}$$

$$n_i = p_i$$

$$I_i \approx 0$$

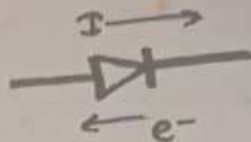
(poor conductor)



Semiconductor Diodes

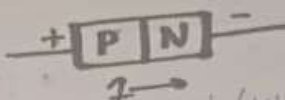
From both n-type and p-type materials, joining the material with majority carriers of electrons to one with majority carriers of holes. There are three cases being discussed.

- (i) No applied Bias ($V = 0V$)
- (ii) Reverse Biased ($V_0 > V_p$)
- (iii) Forward Biased ($V_p > V_0$)



Formation of p-n junction

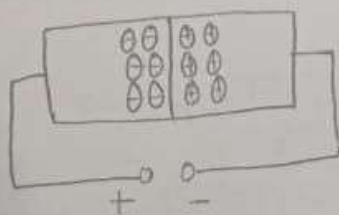
- when an n-type material (with excess e^-) is joined with a p-type material (with excess holes), electrons from the n-side diffuses into p-side and holes from p-side, diffuses into n-side.
- These recombine, leaving behind fixed ions (positively charged donor ions in the n-region and negatively charged acceptor ions in the p-region).
- This forms depletion region, which lacks free charge and acts as an insulating barrier.



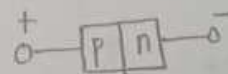
The separation of positive and negative ions creates an internal electric field $\rightarrow$ This field prevents more electrons from diffusing into the p-region and more holes from diffusing into the n-region $\rightarrow$ The voltage associated with this field is called built in potential (V_0).

No Bias Condition

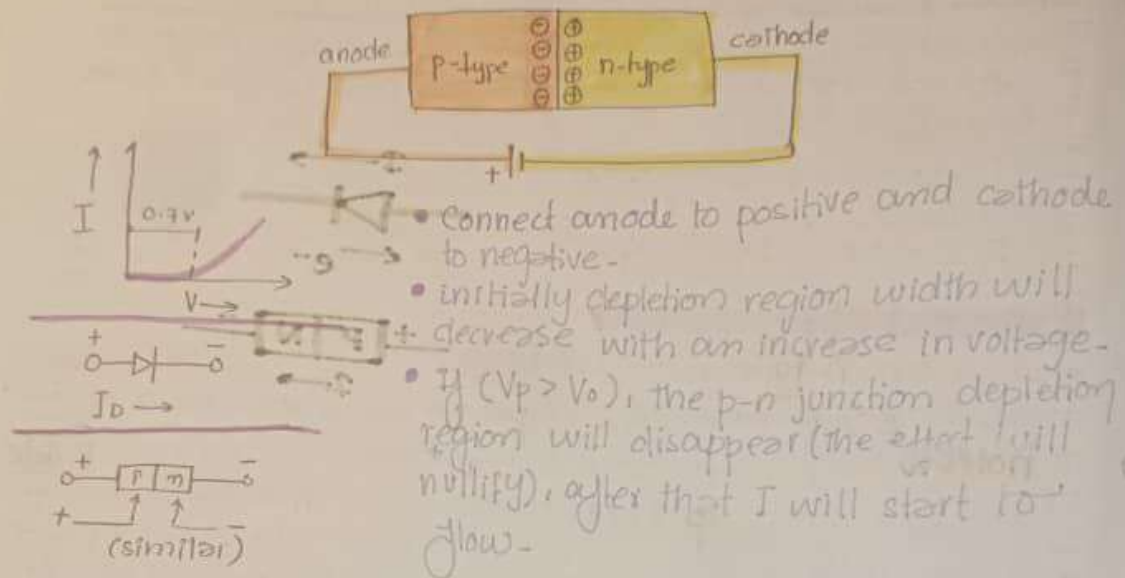
- No external voltage is applied $\rightarrow$ free e^- and holes near the junction combine, leaving only immobile negative and positive ions in depletion region. This region creates an internal electric field preventing further change in carrier movement.



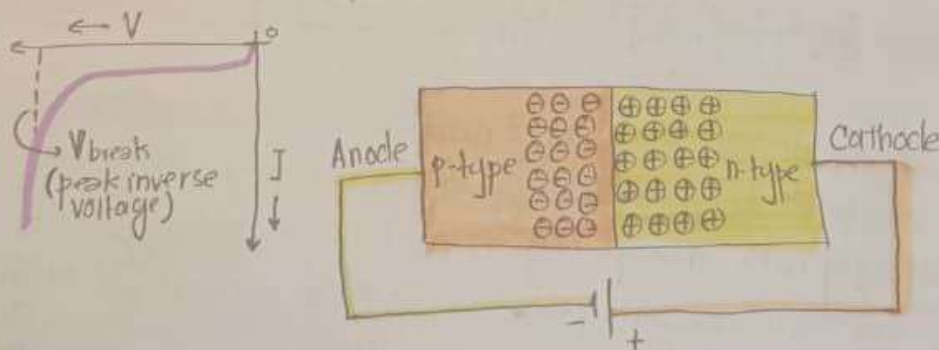
$V_D = 0$
 $I_D = 0mA$



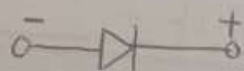
Forward Biased



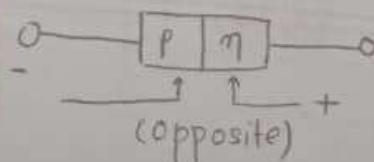
Reverse Biased

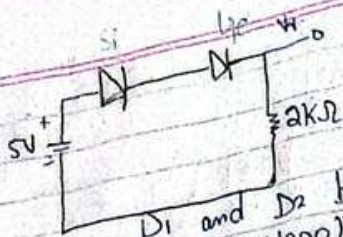


- Connect anode to negative wrt cathode.
- Depletion region width will increase in voltage.
- The reverse voltage does not make any conduct.
- Minor current will flow due to minority carriers.
- After the breakdown voltage, a huge current will flow which will damage the diode.



$I_D \rightarrow I_{sat}$ (Reverse sat. current)





D_1 and D_2 both will conduct

$$-5 + 0.7 + 0.3 + 2(1000)I = 0$$

$$I = \frac{4}{2000} = 2\text{mA}$$

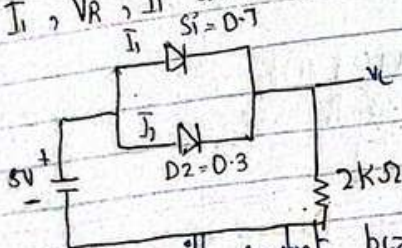
For V_1

$$-2000I + V_1 = 0$$

$$V_1 = 2000(2 \times 10^{-3})$$

$$V_1 = 4\text{V}$$

Find I_1 , V_R , I_1 and I_2



Only D_2 will conduct bcz of low barrier

$$-5 + 0.3 + 2000I = 0$$

$$-4.7 + 2000I = 0$$

$$2000I = 4.7$$

$$I = 2.35\text{mA}$$

For V_1 :

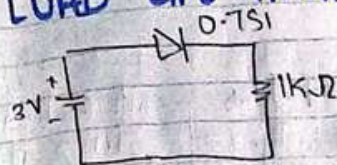
$$-2000I + V_1 = 0$$

$$V_1 = 4.7\text{V}$$

$$I_1 = 0$$

$$I = \frac{V}{R} = \frac{3}{5000}$$

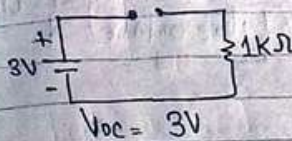
LOAD LIFE ANALYSIS:



Open Circuit $\Rightarrow$ Voltage

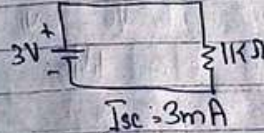
Short Circuit $\Rightarrow$ Current

Open Circuit diode



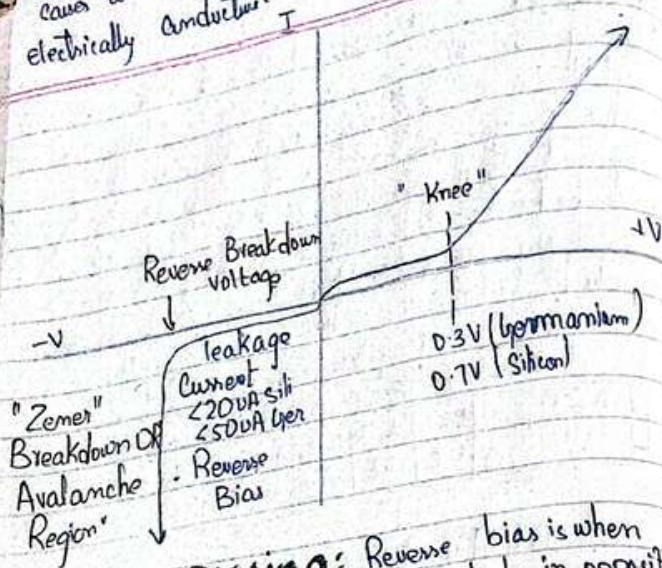
$$V_{oc} = 3\text{V}$$

$$I_{sc} = I_{sc}$$

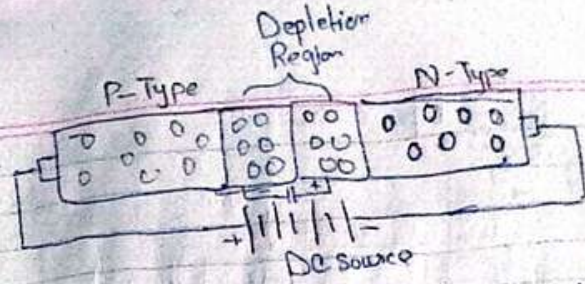


$$I_{sc} = 3\text{mA}$$

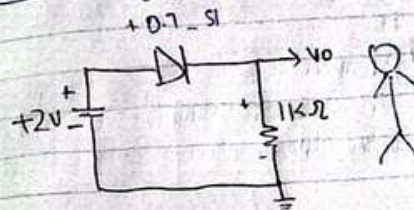
Breakdown voltage is minimum voltage at which an insulating component suddenly becomes electrically conductive.



Reverse Biasing: Reverse bias is when voltage is applied to P-n junction diode in opposite direction of normal current flow, connecting the positive terminal to the n-type material and the negative to the p-type, making diode act like open switch that blocks current by widening the depletion region. A small amount of current flows due to minority carriers, increasing slightly with voltage. If reverse voltage becomes too high, the junction can break down, and current may flow rapidly.



Support drift, drift dominated, $V_{app} < 0$ (-ve)
 $I \approx 0$ (negligible)



find I and V_{out} .

for V_{out}

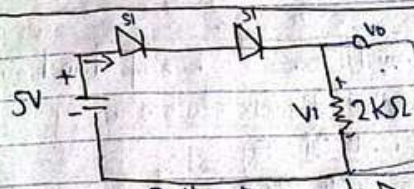
$$-2 + 0.7 + 1000I = 0$$

$$I = 1.3 \text{ mA}$$

$$-1000I + V_0 = 0$$

$$V_0 = 1000I$$

$$V_0 = 1.3 \text{ V}$$



find I and

Both D_1 and D_2 will be on

$$-5 + 0.7 + 0.7 + 2000I = 0$$

$$I = 2.15 \text{ mA}$$

for V_0 ,

$$-2000I + V_0 = 0$$

$$V_0 = 2000(2.15 \times 10^{-3})$$

$$= 4.3 \text{ V}$$

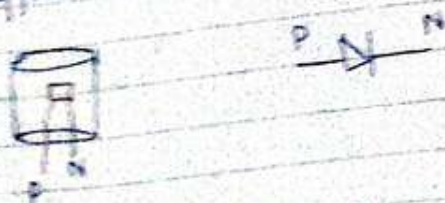
DIODE: a semiconductor device with two terminals, typically allowing the flow of I in one direction.

Two Terminal: Electrically, Silicon diode (mostly used)



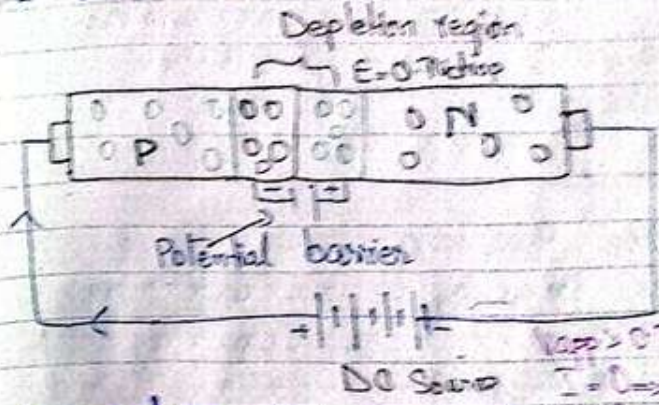
Diffusion: Diffusion in diode is the movement of charge carriers (electrons and holes) from a region of high concentration to a region of low concentration across a p-n junction, creating diffusion current.

Drifting: In a diode, drift refers to directed movement of charge carriers (E and h) due to applied electric field.



BIASING: It is the process of applying a steady DC Voltage or current to an electronic component like transistor or diode, to set its proper operating condition.

Forward Biased: Forward Biasing is applying a voltage across a P-N junction (like in a battery) with positive terminal to the p-type material and negative to the n-type, which opposes the internal potential barrier, shrinks the depletion region, and allows easy current flow from p to n, turning the diode on and will act like closed switch.



Connection: Positive voltage to p-side and negative voltage to n-side (cathode).

Barrier Reduction: The external voltage opposes the diode's internal electric field.

Depletion Region Shrinks: The depletion region becomes thinner and less resistive.

Current flow: Majority carriers overcome barrier and cross the junction, allowing current to flow.

In forward biased $V_{app} > 0V$ (+ve)

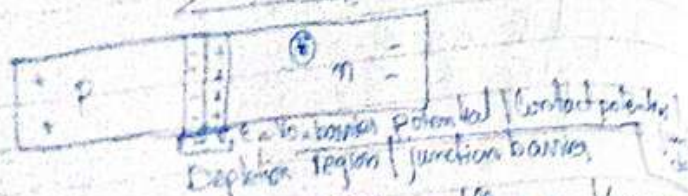
Doping

n-type $\rightarrow e^-$
 P-type $\rightarrow h^+$



Amphoteric $\rightarrow$ for compound semiconductor we can say P or n-type.

$\xrightarrow{h^+}$ diffusion
 $\xleftarrow{e^-}$



The flow due to concentration difference known as diffusion.



field will be opposite of concentration. The flow of charges due to electric field is called drift.

In equilibrium of diode no current will flow.

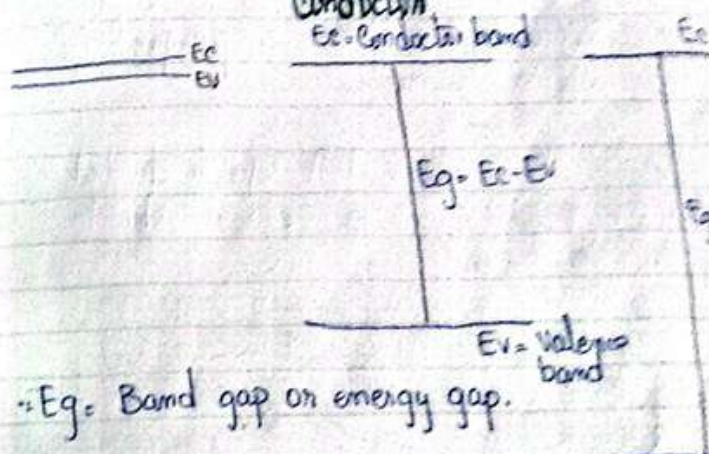
ENERGY BAND GAP:

The amount of energy required to make a semiconductor conductor. It is the energy difference between conductor band and valence band called energy gap.

Conductor

Semi Conductor

Insulator



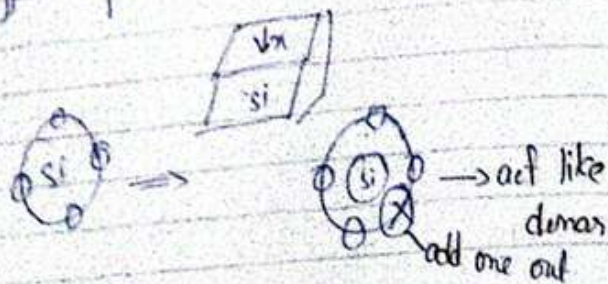
Si → metalloid → grey like glass → not fully transparent

add impurities in Si to get is called doping.

Doping

- 1) n-type
- 2) p-type
- 3) Amphoteric

n-type impurities → donor

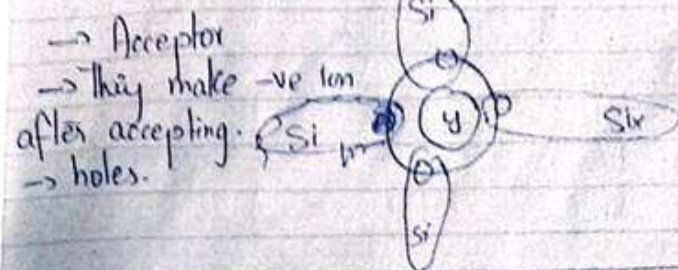


n-type impurities are electron type donor
donor of e^-

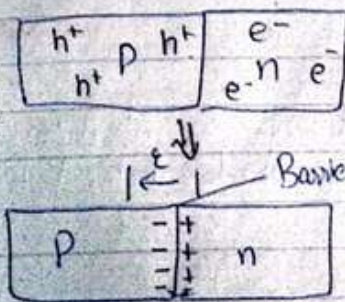
when battery attached it will generate electricity. and this electricity coming from that impurity (n-type doping)

unknown impurity won't provide current

P-Type Impurity:



PN-Junction



The flow of charge due to difference in concentration is called diffusion

pn junction is in equilibrium

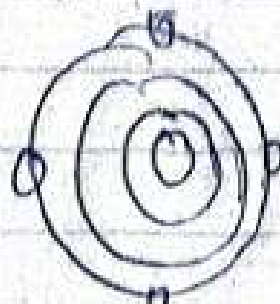
PN Junction diode:



based material $\rightarrow$ silicon $\rightarrow$ semiconductor $\rightarrow$ P.T = 94

Purification

Sand $\rightarrow$ Silicon 94 $\Rightarrow$ At no. 14



pure $\rightarrow$ $1:10^{10}$ 0.8 $1:10^{14}$ $\Rightarrow$ memory chip
 $\Downarrow$ $\Rightarrow$ nano

normal controls, ok to design chip, micro or VLSI chip
size (1 million above to

ultra ASIC $\rightarrow$ application specific IC

Arduino $\rightarrow$ semi custom design

Semi Custom $\rightarrow$ man mani nhi karega